

Library Database - Project Report

CMPT 354 - D100: Database Systems I

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Specifications

A community library needs a database to manage the two sides of its operations. The first side manages literary resources: the items themselves, checkouts, holds, donations, and future acquisitions. The second side manages daily operations: personnel, volunteers, rooms, and events. Library members sit at the intersection: they can use literary items, but they can also attend events, volunteer, and ask librarians for assistance. All literary resources and events are free for members.

Member: A person who uses library resources is a member who is uniquely identified by an ID (Member ID). A member also has a name, address, date of birth, and date of registration. A member need not have any holds, borrows, sign-ups, or donations.

Literary Resources:-

Items: A library has a catalog of literary items that are each uniquely identified by an ID (Item ID). An item also has a name, author, publisher, date of publication, and category type (print book, online book, magazine, scientific journal, records, etc.). An item has one or more copies.

Copies: An item has one or more copies that are each uniquely identified by an ID (Copy ID). A copy has a date of acquisition. A copy can belong to only one item.

Holds: A member (Member ID) can place a hold on any item (Item ID). The member and item pair uniquely identifies the hold, so members cannot place multiple holds on the same item. A hold also has a date of hold and a ready date (initially NULL). A member can have a maximum of ten holds at a given time. A member's hold is removed fourteen days after the ready date.

Borrows: A member (Member ID) can borrow an item copy (Copy ID) from the library where the borrow is uniquely identified by an ID (Borrow ID). A borrow also has a date of checkout, date of return (initially NULL), and an extension amount (0 days, 7 days, 14 days). A copy with a NULL date of return cannot be borrowed by someone else because the copy has not been returned yet. A borrow's due date is calculated as twenty-one days after its checkout date plus its extension. A borrow's fine is calculated as the number of days by which the return date exceeds the due date, multiplied by a constant rate of \$0.25 a day; a copy returned on or before its due date incurs no fine. A borrow's fine does not restrict a member's borrowing or event privileges.

Future Acquisitions: A library keeps a list of items it may add in the future that are each uniquely identified by an ID (Future ID). A future acquisition also has a name, author, publisher, date of publication, category type (print book, online book, magazine, scientific journal, records, etc.), and price. Future acquisitions are tracked independently of the item catalog and are not automatically linked to an Item record.

Donations: A member (Member ID) may donate an item to the library. If the item is not already in the catalog, a new Item and Copy record are created as part of the donation; otherwise, a new Copy record is created for the existing Item. The resulting copy uniquely identifies the donation. A donation also has a date of donation.

Daily Operations Resources:-

Personnel: A library has personnel responsible for day-to-day library operations (assisting members, answering questions, etc.) who are uniquely identified by an ID (Employee ID). An employee also has a name, address, department, job title, salary, and date of hire. An employee's job title is independent of their salary.

Volunteers: A library member (Member ID) can also volunteer at the library. A volunteer is already uniquely identified by their Member ID.

Rooms: A library has social rooms that are each uniquely identified by an ID (Room ID). A room also has a maximum capacity and a description of the equipment in it (projector, sound, etc.). This equipment information helps staff assign a suitable room to each event.

Events: A library holds events that are each uniquely identified by an ID (Event ID). An event also has a title, room (Room ID), date, start time, end time, maximum capacity, type (book club, book-related event, art show, film screening, etc.), and a recommended minimum audience age. An event takes place in exactly one room and two events cannot be scheduled in the same room at overlapping times. A room's maximum capacity must be greater than or equal to the event's maximum capacity. An event's type is independent of its minimum age.

Sign-Ups: A member (Member ID) can sign up to attend an event (Event ID). The member and event pair uniquely identifies the sign-up. A sign-up also has a date of sign-up. A member cannot sign up for an event that has already reached its maximum capacity.

Event Staffing: An employee (Employee ID) can be assigned to an event (Event ID). The event and employee pair uniquely identifies the event staffing record.

Event Volunteering: A volunteer (Member ID) can be assigned to an event (Event ID). The event and volunteer pair uniquely identifies the event volunteering record. An event volunteering record also has an assigned role for the volunteer (can be blank).

Assistance Request: A member (Member ID) can request assistance where the request is uniquely identified by an ID (Assist ID). A request also has a requested librarian (Employee ID), request text, and date of submission.

E/R Diagram

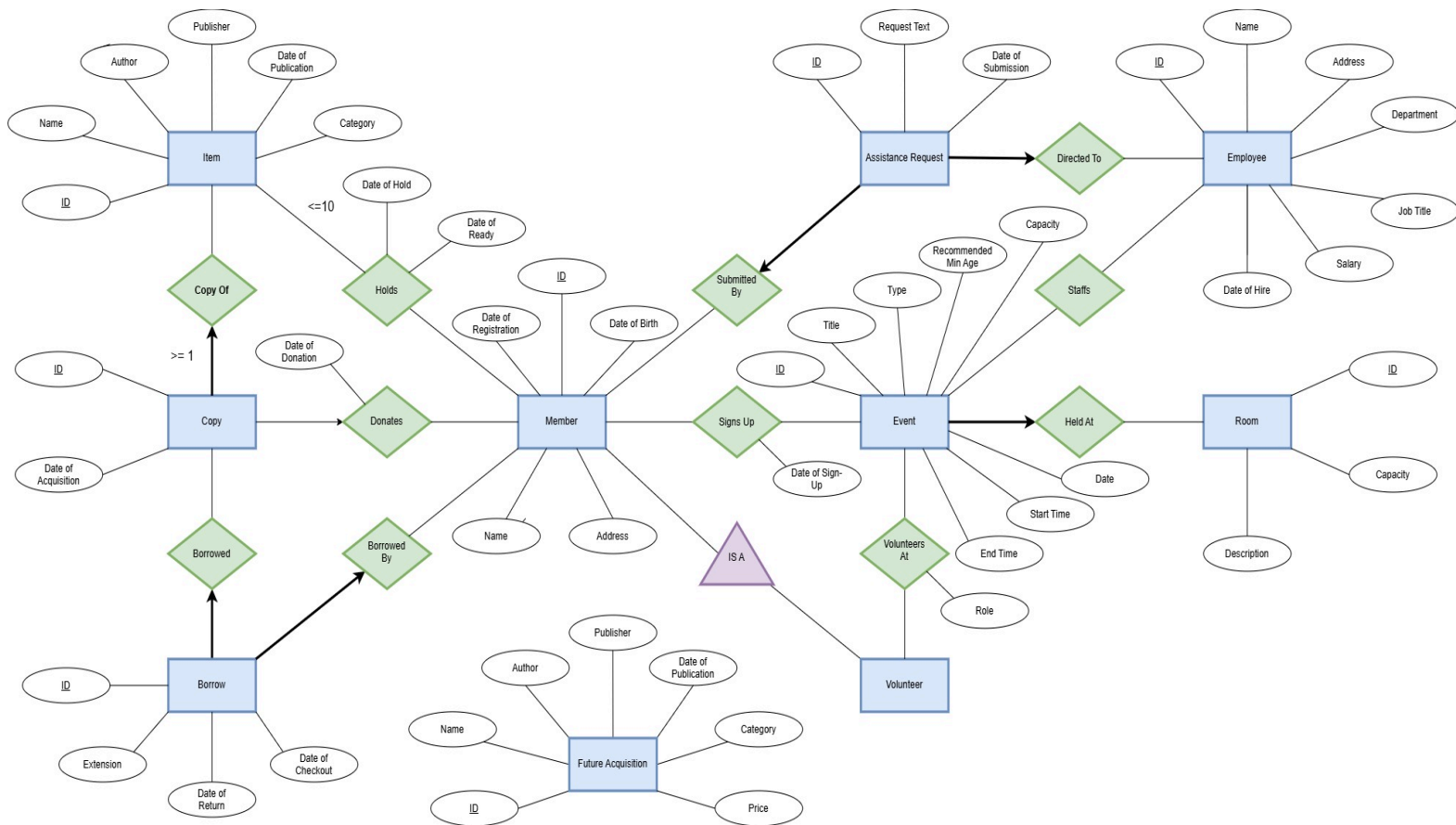


Figure 1: E/R Diagram

Anomaly Analysis

BCNF rules out FD anomalies by definition: anomalies require redundancy, redundancy requires a non-superkey determinant, and BCNF forces every non-trivial FD determinant to be a superkey. Reversing the chain, every determinant is a superkey, so the schema cannot have the redundancies that lead to insertion, update, and deletion anomalies.

The table below lists each relation's non-trivial functional dependencies (from the specification in Step 1), the candidate keys derived from them, and their BCNF verification. Every determinant listed is a superkey, so the design is already in BCNF and contains no bad functional dependencies. No further decomposition is needed beyond what is outlined in Steps 1 and 2. The relation schemas this design translates to are listed below the table.

Relation	Functional Dependencies	Keys	BCNF
Members	Member ID $\rightarrow$ Name, Address, Date of Birth, Date of Registration	{Member ID}	Yes, the only non-trivial FD determinant is the key {Member ID}
Items	Item ID $\rightarrow$ Name, Author, Publisher, Date of Publication, Category	{Item ID}	Yes, the only non-trivial FD determinant is the key {Item ID}
Copies	Copy ID $\rightarrow$ Item ID, Date of Acquisition	{Copy ID}	Yes, the only non-trivial FD determinant is the key {Copy ID}
Holds	Member ID, Item ID $\rightarrow$ Date of Hold, Date of Ready	{Member ID, Item ID}	Yes, the only non-trivial FD determinant is the key {Member ID, Item ID}
Borrows	Borrow ID $\rightarrow$ Member ID, Copy ID, Date of Checkout, Date of Return, Extension	{Borrow ID}	Yes, the only non-trivial FD determinant is the key {Borrow ID}
Future Acquisitions	Future ID $\rightarrow$ Name, Author, Publisher, Date of Publication, Category, Price	{Future ID}	Yes, the only non-trivial FD determinant is the key {Future ID}
Donations	Copy ID $\rightarrow$ Member ID, Date of Donation	{Copy ID}	Yes, the only non-trivial FD determinant is the key {Copy ID}

Employees	Employee ID $\rightarrow$ Name, Address, Department, Job Title, Salary, Date of Hire	{Employee ID}	Yes, the only non-trivial FD determinant is the key {Employee ID}
Volunteers	No non-trivial functional dependencies. Member ID is the key and there are no other attributes.	{Member ID}	Yes, there are no non-trivial FDs, the relation is all-key {Member ID}
Rooms	Room ID $\rightarrow$ Capacity, Equipment Description	{Room ID}	Yes, the only non-trivial FD determinant is the key {Room ID}
Events	Event ID $\rightarrow$ Title, Room ID, Date, Start Time, End Time, Capacity, Type, Recommended Min Age Room ID, Date, Start Time $\rightarrow$ Event ID Room ID, Date, End Time $\rightarrow$ Event ID	{Event ID} {Room ID, Date, Start Time} {Room ID, Date, End Time}	Yes, all three non-trivial FD determinants are candidate keys: {Event ID} {Room ID, Date, Start Time} {Room ID, Date, End Time}
Sign-Ups	Event ID, Member ID $\rightarrow$ Date of Signup	{Event ID, Member ID}	Yes, the only non-trivial FD determinant is the key {Event ID, Member ID}
Event Staffing	No non-trivial functional dependencies. Employee ID and Event ID are the key and there are no other attributes.	{Event ID, Employee ID}	Yes, there are no non-trivial FDs, the relation is all-key {Event ID, Employee ID}
Event Volunteering	Event ID, Member ID $\rightarrow$ Role	{Event ID, Member ID}	Yes, the only non-trivial FD determinant is the key {Event ID, Member ID}
Assistance Request	Assist ID $\rightarrow$ Member ID, Employee ID, Request Text, Date of Submission	{Assist ID}	Yes, the only non-trivial FD determinant is the key {Assist ID}

Table 1: Functional Dependencies, Keys, BCNF

Relation Schemas

- Members (MemberID, Name, Address, Date of Birth, Date of Registration)

- Items (ItemID, Name, Author, Publisher, Date of Publication, Category)
- Copies (CopyID, ItemID, Date of Acquisition)
- Holds (MemberID, ItemID, Date of Hold, Date of Ready)
- Borrows (BorrowID, MemberID, CopyID, Date of Checkout, Date of Return, Extension)
- Future Acquisitions (FutureID, Name, Author, Publisher, Date of Publication, Category, Price)
- Donations (CopyID, MemberID, Date of Donation)
- Employees (EmployeeID, Name, Address, Department, Job Title, Salary, Date of Hire)
- Volunteers (MemberID)
- Rooms (RoomID, Capacity, Equipment Description)
- Events (EventID, Title, RoomID, Date, Start Time, End Time, Capacity, Type, Recommended Min Age)
- Sign-Ups (EventID, MemberID, Date of Signup)
- Event Staffing (EventID, EmployeeID)
- Event Volunteering (EventID, MemberID, Role)
- Assistance Requests(AssistID, MemberID, EmployeeID, Request Text, Date of Submission)

Rule	Implementation
The date of birth must be less than the date of registration.	CHECK on Members
The category must only be one of the listed category types.	CHECK on Items, Future Acquisitions
The price must be greater than 0.	CHECK on Future Acquisitions
The salary must be greater than 0.	CHECK on Employees
The date of ready must be greater than or equal to the date of hold.	CHECK on Holds
The extension must be 0, 7, or 14 days.	CHECK on Borrows
The date of return must be greater than or equal to the date of checkout.	CHECK on Borrows
The end time must be greater than the start time.	CHECK on Events
The capacity must be greater than 0.	CHECK on Rooms, Events

The date and time must be in correct form.	CHECK on Members, Items, Copies, Holds, Borrows, Future Acquisitions, Donations, Employees, Events, Sign-Ups, Assistance Requests
Members must not have more than 10 holds.	TRIGGER
Unreturned copies must not be checked out.	TRIGGER
Items must have at least one copy.	TRIGGER
Events must not overlap in time and room.	TRIGGER
The room capacity must be greater than or equal to the event capacity.	TRIGGER
The sign-up count cannot exceed the event capacity.	TRIGGER

Table 2: Constraints, Triggers

Enforcement

Views: The due date is determined by the date of checkout and extension. The fine is determined by the date of checkout, the date of return, and extension. Since neither determinant is a superkey, storing the attribute would violate BCNF. Thus, the values of due date and fines are derived on the fly with a VIEW on Borrow.

Applications: The application ensures that for each donation a new item record is created for the copy if necessary (by comparing attributes), adds a copy record, and then finally adds a donation record. The application ensures that expired holds are removed fourteen days after their ready date when reconnecting to the database.